Mangroves and Coastal Ecosystems

Knowledge document 040 of 100

Introduction to Mangroves

Mangroves are salt-tolerant trees and shrubs that grow in the intertidal zone of tropical and subtropical coasts, in muddy or sandy areas where rivers meet the sea, such as estuaries, deltas, lagoons and backwaters. Their habitat is flooded by sea water at high tide and exposed at low tide, with salty, oxygen-poor mud. The word mangrove refers both to the individual plants and to the whole ecosystem, which is also called a mangrove forest or swamp. Mangroves grow in more than 120 countries and cover about 147,000 square kilometres worldwide, with the largest areas in Indonesia, Brazil, Nigeria, Mexico and Australia. They are among the most productive and valuable ecosystems on Earth.

How Mangroves Survive: Special Adaptations

Mangroves have developed remarkable adaptations. Stilt roots or prop roots, as in Rhizophora, grow from the trunk and branches into the mud to hold the tree firmly against waves. Pneumatophores, or breathing roots, as in Avicennia, stick up above the mud like pencils and take in oxygen for roots buried in oxygen-poor mud. Many species have salt-excreting glands in their leaves, or filter out salt at the roots, to cope with salt water. Their seeds often germinate while still attached to the parent tree, a trait called vivipary, and the young seedlings, called propagules, drop into the water and float until they find soil. Common mangrove species include the red mangrove (Rhizophora), black mangrove (Avicennia), white mangrove and Sonneratia, Bruguiera and Excoecaria.

Where Is the Pichavaram Mangrove Forest?

The Pichavaram mangrove forest is located in Cuddalore district of Tamil Nadu, near the town of Chidambaram and about 200 kilometres south of Chennai, on the Bay of Bengal coast. It lies between the Vellar estuary in the north and the Coleroon (Kollidam) estuary in the south, where fresh river water mixes with sea water. It covers roughly 1,100 hectares and is one of the largest mangrove forests in India. It consists of a maze of small islands, creeks and canals, with dense stands of Rhizophora, Avicennia and other species, more than 50 types of birds, fish, crabs and prawns, and it attracts many tourists for boat rides and bird-watching. Winter brings migratory birds. Pichavaram has appeared in many Tamil and other Indian films because of its scenic beauty. Another important mangrove site in Tamil Nadu is Muthupet in Tiruvarur and Thanjavur districts, and others lie at Ramanathapuram, Kazhuveli and the Adyar and Ennore estuaries in Chennai.

How Do Mangroves Protect Coasts?

Mangroves form a natural shield for the coast. Their dense network of roots and trunks slows down and absorbs the energy of waves, storm surges, cyclones and tsunamis, reducing damage to land behind them. After the Indian Ocean tsunami of December 2004, studies and reports found that villages sheltered by thick mangroves, such as those near Pichavaram, suffered less damage and loss of life than unprotected areas. The roots also trap sediment and bind soil, which prevents coastal erosion and helps build up land. They filter pollutants and nutrients from the water flowing from the land to the sea, protecting coral reefs and seagrass beds. Mangroves are nurseries for fish, shrimp and crabs, so they support fisheries and the livelihoods of coastal communities, and they provide honey, firewood and tannin. Birds, crocodiles, otters, monitor lizards and in the Sundarbans even Bengal tigers live in them.

Mangroves and Climate: Blue Carbon

Mangroves are very efficient at capturing carbon dioxide, and store it in their biomass and especially in the waterlogged mud, in what is called blue carbon. Per hectare, mangroves can store several times more carbon than most tropical forests, so protecting them helps fight climate change. When mangroves are destroyed, the stored carbon is released.

Mangroves in India

India has about 4,900 to 5,000 square kilometres of mangrove cover, about 0.15 percent of the country's area, according to the India State of Forest Report. The largest mangrove forest in the world is the Sundarbans in the Ganges-Brahmaputra delta, shared between India and Bangladesh, a UNESCO World Heritage Site since 1987, famous for the Royal Bengal tiger. West Bengal, Gujarat and the Andaman and Nicobar Islands have the most mangroves in India. The Bhitarkanika mangroves in Odisha are known for saltwater crocodiles and Olive Ridley turtles. India launched the MISHTI scheme (Mangrove Initiative for Shoreline Habitats and Tangible Incomes) in 2023 to expand mangrove cover. The International Day for the Conservation of the Mangrove Ecosystem is observed on 26 July, designated by UNESCO in 2015.

Other Coastal Ecosystems

Mangroves work together with other coastal ecosystems. Coral reefs, such as those in the Gulf of Mannar in Tamil Nadu, shelter many fish species and break the waves. Seagrass meadows, as in the Palk Bay, feed dugongs, and salt marshes, estuaries and sand dunes also protect the coast. The Gulf of Mannar Biosphere Reserve, India's first marine biosphere reserve, was set up in 1989.

Threats and Conservation

Roughly one-fifth of the world's mangroves have been lost since 1980 because of conversion to shrimp farms, agriculture, urban development, ports, pollution, oil spills, sand mining and sea-level rise. Mangroves are protected in India by the Coastal Regulation Zone notification, which treats them as ecologically sensitive areas, and by community-based restoration programmes.

Challenges and Future

Challenges include coastal development, reduced freshwater flow due to dams, and climate change. Restoration projects, including community planting in Tamil Nadu, and recognition of the value of blue carbon, give hope for a better future for these valuable coastal forests.

Quick Facts

- Mangroves are salt-tolerant trees and shrubs growing in intertidal coastal zones.
- Pichavaram mangrove forest lies in Cuddalore district, Tamil Nadu, near Chidambaram.
- Mangroves protect coasts by absorbing wave energy, trapping sediment and reducing storm and tsunami damage.
- The Sundarbans is the world's largest mangrove forest and a UNESCO site since 1987.
- International Day for the Conservation of the Mangrove Ecosystem is on 26 July.